

Week 1 Class Activity

Apurva Timilsina

Westcliff University

PRG 310

Professor Mathema

March 13, 2026

Week 1 Class Activity

DevOps has been implemented by many large technology companies to enhance the speed and reliability of software development, in addition to collaboration. The most recognizable ones are Etsy, Netflix, and Amazon. Such companies had a number of challenges in the process of adoption but also had substantial improvements.

1. Netflix

The problem of managing the high growth rates and the reliability of the system to millions of users across the globe was one of the significant problems Netflix had to solve. The platform had to increase the infrastructure scale and high-availability of the streaming services as it grew. To handle this, the practices of DevOps that Netflix adopted include automated deployment pipelines, microservices architecture, and continuous integration/continuous delivery (CI/CD). There were also tools such as chaos engineering that were implemented to test the resilience of systems through intentional failures. Consequently, Netflix had more rapid deployments and better system availability with thousands of production updates daily and with high uptime (DevOps Support, 2024).

2. Amazon

Amazon had a problem with having thousands of services managed and regular updates in its huge infrastructure. Traditional approaches to development slowed development and decreased flexibility. Amazon applied the DevOps principles of you build it you run it to have the developers responsible for their code, which is developed through to production. This cultural transformation enhanced ownership, quality of code and the recovery time. Thousand performances a day afterward, continuous delivery pipelines and infrastructure automation also allowed Amazon to do thousands of deployments a day (NASSCOM, 2024).

3. Etsy

Manual releases and poor workflows caused slow deployment and frequent outages at etsy. Once DevOps practices such as automated tests, CI/CD pipelines, and configuration management tools have been implemented, Etsy has radically changed the development workflow. The number of deployments per day rose to over 50 and the downtime decreased, making the company more productive in terms of developers (DevOps Consulting, 2024).

Overall, these case studies show that despite the difficulty of DevOps adoption, it significantly increases the speed of deployment, system reliability, and collaboration in the development teams.

References

DevOps Consulting. (2024). Top 10 real-world DevOps implementation case studies.

DevOps Support. (2024). Real-world examples of successful DevOps implementation.

NASSCOM. (2024). Building a DevOps culture: Challenges and success stories.